Overview

In our examination of wave propagation on a transmission line in Chapter 2, the primary quantities we worked with were voltage, current, impedance, and power. Each of these is a ***scalar*** quantity, meaning that it can be completely specified by its magnitude if it is a positive real number or by its magnitude and phase angle if it is a negative or a complex number (a negative number has a positive magnitude and a phase angle of π (rad)). This chapter is concerned with vectors. A ***vector*** has a magnitude and a direction. The speed of an object is a scalar, whereas its velocity is a vector.

Starting in the next chapter and throughout the succeeding chapters in this book, the primary electromagnetic quantities we deal with are the electric and magnetic fields, **E** and **H**. These, and many other related quantities, are vectors. Vector analysis provides the mathematical tools necessary for expressing and manipulating vector quantities in an efficient and convenient manner. To specify a vector in three-dimensional space, it is necessary to specify its components along each of the three directions.

► Several types of coordinate systems are used in the study of vector quantities, the most common being the Cartesian (or rectangular), cylindrical, and spherical systems. A particular coordinate system is usually chosen to best suit the geometry of the problem under consideration. ◄

Vector algebra governs the laws of addition, subtraction, and "multiplication" of vectors. The rules of vector algebra and vector representation in each of the aforementioned orthogonal coordinate systems (including vector transformation between them) are two of the three major topics treated in this chapter. The third topic is ***vector calculus***, which encompasses the laws of differentiation and integration of vectors, the use of special vector operators (gradient, divergence, and curl), and the application of certain theorems that are particularly useful in the study of electromagnetics, most notably the divergence and Stokes's theorems.

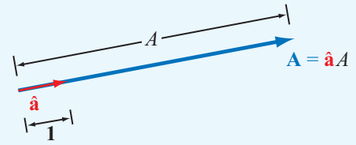
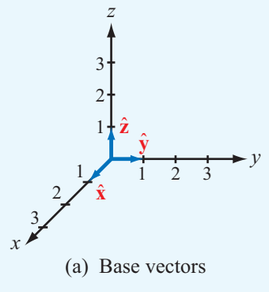
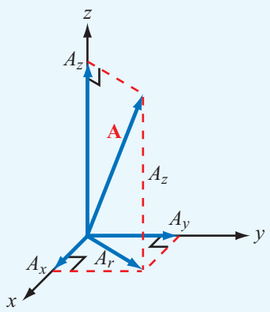


Figure 3-1 Vector $\mathbf{A} = \hat{\mathbf{a}}A$ has magnitude $A = |\mathbf{A}|$ and points in the direction of unit vector $\hat{\mathbf{a}} = \mathbf{A}/A$.



(a) Base vectors



(b) Components of A

Figure 3-2 Cartesian coordinate system: (a) base vectors $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$ and (b) components of vector **A**.

3-1 Basic Laws of Vector Algebra

A vector is a mathematical object that resembles an arrow. Vector **A** in **Fig. 3-1** has ***magnitude*** (or length) $A = |\mathbf{A}|$ and ***unit vector*** $\hat{\mathbf{a}}$:

$$\mathbf{A} = \hat{\mathbf{a}}|\mathbf{A}| = \hat{\mathbf{a}}A. \tag{3.1}$$

The unit vector $\hat{\mathbf{a}}$ has a magnitude of one ($|\hat{\mathbf{a}}| = 1$) and points from **A**'s tail or anchor to its head or tip. From Eq. (3.1),

$$\hat{\mathbf{a}} = \frac{\mathbf{A}}{|\mathbf{A}|} = \frac{\mathbf{A}}{A}. \tag{3.2}$$

In the Cartesian (or rectangular) coordinate system shown in **Fig. 3-2(a)**, the x, y, and z coordinate axes extend along

概述

在第 2 章对传输线上的波传播的研究中，我们使用的主要量是电压、电流、阻抗和功率。其中每一个都是标量，这意味着如果它是正实数，则可以完全通过其大小来指定；如果它是负数或复数，则可以通过其大小和相位角来完全指定（负数具有正大小和 π (rad) 的相位角）。本章涉及向量。向量具有大小和方向。物体的速度是标量，而速度是矢量。

从下一章开始以及本书的后续章节，我们处理的主要电磁量是电场和磁场，E和H。这些以及许多其他相关量都是矢量。矢量分析提供了以高效、便捷的方式表达和操作矢量所需的数学工具。要在三维空间中指定一个向量，必须指定其沿三个方向的分量。

► 在矢量研究中使用了几种类型的坐标系，最常见的是笛卡尔（或矩形）坐标系、圆柱坐标系和球坐标系。通常选择一个特定的坐标系来最适合所考虑问题的几何形状。 ◄

向量代数控制向量的加法、减法和“乘法”定律。上述每个正交坐标系中的向量代数和向量表示（包括它们之间的向量变换）的规则是本章讨论的三个主要主题中的两个。第三个主题是向量微积分，其中包括向量的微分和积分定律、特殊向量算子（梯度、散度和旋度）的使用，以及某些在电磁学研究中特别有用的定理的应用，尤其是散度和斯托克斯定理。

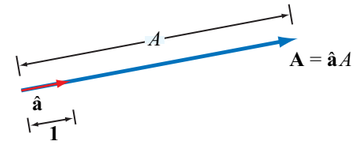
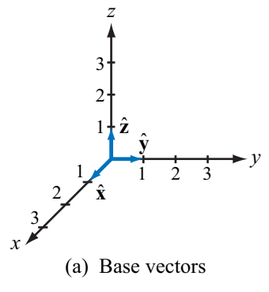
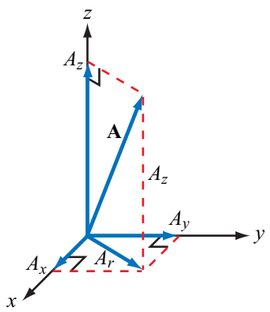


图 3-1 向量 A = âA 的大小为 A = |A|，并指向单位向量 â = A/A 的方向。



(a) Base vectors



(b) Components of A

图 3-2 笛卡尔坐标系：(a) 基向量 x̂、ŷ 和 ẑ 以及 (b) 向量 A 的分量。

3-1 向量代数的基本定律

矢量是一个类似于箭头的数学对象。图 3-1 中的向量 A 具有大小（或长度）A = |A| 和单位向量 â：

$$\mathbf{A} = \hat{\mathbf{a}}|\mathbf{A}| = \hat{\mathbf{a}}A. \tag{3.1}$$

单位向量 â 的大小为 1 ($|\hat{\mathbf{a}}| = 1$)，并从 A 的尾部或锚点指向其头部或尖端。从方程。 (3.1),

$$\hat{\mathbf{a}} = \frac{\mathbf{A}}{|\mathbf{A}|} = \frac{\mathbf{A}}{A}. \tag{3.2}$$

在图3-2(a)所示的笛卡尔（或直角）坐标系中，x、y、z 坐标轴沿

directions of the three mutually perpendicular unit vectors $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$, which are also called ***base vectors***. The vector $\mathbf{A}$ in **Fig. 3-2(b)** may be decomposed as

$$\mathbf{A} = \hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z, \tag{3.3}$$

where A_x, A_y, and A_z are $\mathbf{A}$'s scalar components along the x-, y-, and z axes, respectively. The component A_z is equal to the perpendicular projection of $\mathbf{A}$ onto the z axis, and similar definitions apply to A_x and A_y. Application of the Pythagorean theorem—first to the right triangle in the x–y plane to express the hypotenuse A_r in terms of A_x and A_y and then again to the vertical right triangle with sides A_r and A_z and hypotenuse A—yields the following expression for the magnitude of $\mathbf{A}$:

$$A = |\mathbf{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}. \tag{3.4}$$

Since A is a nonnegative scalar, only the positive root applies. From Eq. (3.2), the unit vector $\hat{\mathbf{a}}$ is

$$\hat{\mathbf{a}} = \frac{\mathbf{A}}{A} = \frac{\hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z}{\sqrt{A_x^2 + A_y^2 + A_z^2}}. \tag{3.5}$$

Occasionally, we use the shorthand notation $\mathbf{A} = (A_x, A_y, A_z)$ to denote a vector with components A_x, A_y, and A_z in a Cartesian coordinate system.

3-1.1 Equality of Two Vectors

Two vectors $\mathbf{A}$ and $\mathbf{B}$ are equal if they have equal magnitudes and identical unit vectors. Thus, if

$$\mathbf{A} = \hat{\mathbf{a}}A = \hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z, \tag{3.6a}$$

$$\mathbf{B} = \hat{\mathbf{b}}B = \hat{\mathbf{x}}B_x + \hat{\mathbf{y}}B_y + \hat{\mathbf{z}}B_z, \tag{3.6b}$$

then $\mathbf{A} = \mathbf{B}$ if and only if $A = B$ and $\hat{\mathbf{a}} = \hat{\mathbf{b}}$, which requires that $A_x = B_x$, $A_y = B_y$, and $A_z = B_z$.

▶ Equality of two vectors does not necessarily imply that they are identical; in Cartesian coordinates, two displaced parallel vectors of equal magnitude and pointing in the same direction are equal, but they are identical only if they lie on top of one another. ◀

3-1.2 Vector Addition and Subtraction

The sum of two vectors $\mathbf{A}$ and $\mathbf{B}$ is a vector

$$\mathbf{C} = \hat{\mathbf{x}}C_x + \hat{\mathbf{y}}C_y + \hat{\mathbf{z}}C_z,$$



(a) Parallelogram rule (b) Head-to-tail rule

Figure 3-3 Vector addition by (a) the parallelogram rule and (b) the head-to-tail rule.

given by

$$\begin{aligned}\mathbf{C} = \mathbf{A} + \mathbf{B} &= (\hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z) + (\hat{\mathbf{x}}B_x + \hat{\mathbf{y}}B_y + \hat{\mathbf{z}}B_z)\\ &= \hat{\mathbf{x}}(A_x + B_x) + \hat{\mathbf{y}}(A_y + B_y) + \hat{\mathbf{z}}(A_z + B_z)\\ &= \hat{\mathbf{x}}C_x + \hat{\mathbf{y}}C_y + \hat{\mathbf{z}}C_z,\end{aligned} \tag{3.7}$$

with $C_x = A_x + B_x$, etc.

▶ Vector addition is commutative:

$$\mathbf{C} = \mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}. \tag{3.8}$$

Graphically, vector addition can be accomplished by either the parallelogram or the head-to-tail rule (**Fig. 3-3**). Vector $\mathbf{C}$ is the diagonal of the parallelogram with sides $\mathbf{A}$ and $\mathbf{B}$. With the head-to-tail rule, we may either add $\mathbf{A}$ to $\mathbf{B}$ or $\mathbf{B}$ to $\mathbf{A}$. When $\mathbf{A}$ is added to $\mathbf{B}$, it is repositioned so that its tail starts at the tip of $\mathbf{B}$ while keeping its length and direction unchanged. The sum vector $\mathbf{C}$ starts at the tail of $\mathbf{B}$ and ends at the tip of $\mathbf{A}$.

Subtraction of vector $\mathbf{B}$ from vector $\mathbf{A}$ is equivalent to the addition of $\mathbf{A}$ to negative $\mathbf{B}$. Thus,

$$\begin{aligned}\mathbf{D} = \mathbf{A} - \mathbf{B} &= \mathbf{A} + (-\mathbf{B})\\ &= \hat{\mathbf{x}}(A_x - B_x) + \hat{\mathbf{y}}(A_y - B_y) + \hat{\mathbf{z}}(A_z - B_z).\end{aligned} \tag{3.9}$$

Graphically, the same rules used for vector addition are also applicable to vector subtraction; the only difference is that the arrowhead of $(-\mathbf{B})$ is drawn on the opposite end of the line segment representing the vector $\mathbf{B}$ (i.e., the tail and head are interchanged).

3-1.3 Position and Distance Vectors

The ***position vector*** of a point P in space is the vector from the origin to P. Assuming points P_1 and P_2 are at (x_1, y_1, z_1) and (x_2, y_2, z_2) in **Fig. 3-4**, their position vectors are

$$\mathbf{R}_1 = \overrightarrow{OP_1} = \hat{\mathbf{x}}x_1 + \hat{\mathbf{y}}y_1 + \hat{\mathbf{z}}z_1, \tag{3.10a}$$

三个相互垂直的单位向量x̂、ŷ和ẑ的方向，也称为基向量。图3-2(b)中的向量A可分解为

$$\mathbf{A} = \hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z, \tag{3.3}$$

其中 A_x、A_y 和 A_z 分别是 A 沿 x、y 和 z 轴的标量分量。分量 A_z 等于 A 在 z 轴上的垂直投影，类似的定义适用于 A_x 和 A_y。应用毕达哥拉斯定理（首先应用到 x-y 平面中的直角三角形，以 A_x 和 A_y 表示斜边 A_r，然后再次应用到边为 A_r 和 A_z 以及斜边 A 的垂直直角三角形，得出 A 大小的以下表达式：

$$A = |\mathbf{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}. \tag{3.4}$$

由于 A 是非负标量，因此仅适用正根。从方程。 (3.2) 则单位向量â为

$$\hat{\mathbf{a}} = \frac{\mathbf{A}}{A} = \frac{\hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z}{\sqrt{A_x^2 + A_y^2 + A_z^2}}. \tag{3.5}$$

有时，我们使用简写符号 $A = (A_x、A_y、A_z)$ 来表示笛卡尔坐标系中具有分量 A_x、A_y 和 A_z 的向量。

3-1.1 两个向量相等

如果两个向量 A 和 B 具有相等的大小和相同的单位向量，则它们相等。因此，如果

$$\mathbf{A} = \hat{\mathbf{a}}A = \hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z, \tag{3.6a}$$

$$\mathbf{B} = \hat{\mathbf{b}}B = \hat{\mathbf{x}}B_x + \hat{\mathbf{y}}B_y + \hat{\mathbf{z}}B_z, \tag{3.6b}$$

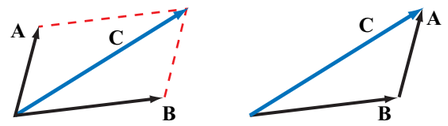
则 $A = B$ 当且仅当 $A = B$ 和 $\hat{a} = \hat{b}$，这要求 $A_x = B_x$、$A_y = B_y$ 和 $A_z = B_z$。

▶ 两个向量相等并不一定意味着它们相同；在笛卡尔坐标系中，两个大小相等且指向同一方向的位移平行向量相等，但仅当它们彼此重叠时才相同。 ◀

3-1.2 向量加法和减法

两个向量 A 和 B 的和是一个向量

$$\mathbf{C} = \hat{\mathbf{x}}C_x + \hat{\mathbf{y}}C_y + \hat{\mathbf{z}}C_z,$$



(a) Parallelogram rule (b) Head-to-tail rule

图 3-3 通过 (a) 平行四边形规则和 (b) 头尾规则进行向量加法。

给出的

$$\begin{aligned}\mathbf{C} = \mathbf{A} + \mathbf{B} &= (\hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z) + (\hat{\mathbf{x}}B_x + \hat{\mathbf{y}}B_y + \hat{\mathbf{z}}B_z)\\ &= \hat{\mathbf{x}}(A_x + B_x) + \hat{\mathbf{y}}(A_y + B_y) + \hat{\mathbf{z}}(A_z + B_z)\\ &= \hat{\mathbf{x}}C_x + \hat{\mathbf{y}}C_y + \hat{\mathbf{z}}C_z,\end{aligned} \tag{3.7}$$

与 $C_x = A_x + B_x$ 等

▶ 向量加法是可交换的：

$$\mathbf{C} = \mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}. \tag{3.8}$$

从图形上看，向量加法可以通过平行四边形或头尾规则来完成（图3-3）。向量 C 是边为 A 和 B 的平行四边形的对角线。根据头尾规则，我们可以将 A 添加到 B 或将 B 添加到 A。当 A 添加到 B 时，它会重新定位，使其尾部从 B 的尖端开始，同时保持其长度和方向不变。和向量 C 从 B 的尾部开始，到 A 的尖端结束。

从向量 A 中减去向量 B 相当于将 A 与负 B 相加。因此，

$$\begin{aligned}\mathbf{D} = \mathbf{A} - \mathbf{B} &= \mathbf{A} + (-\mathbf{B})\\ &= \hat{\mathbf{x}}(A_x - B_x) + \hat{\mathbf{y}}(A_y - B_y) + \hat{\mathbf{z}}(A_z - B_z).\end{aligned} \tag{3.9}$$

从图形上看，用于矢量加法的相同规则也适用于矢量减法；唯一的区别是 $(-B)$ 的箭头画在表示向量 B 的线段的另一端（即尾部和头部互换）。

3-1.3 位置和距离向量

空间中点 P 的位置向量是从原点到 P 的向量。假设图 3-4 中点 P_1 和 P_2 位于 (x_1, y_1, z_1) 和 (x_2, y_2, z_2)，则它们的位置向量为

$$\mathbf{R}_1 = \overrightarrow{OP_1} = \hat{\mathbf{x}}x_1 + \hat{\mathbf{y}}y_1 + \hat{\mathbf{z}}z_1, \tag{3.10a}$$

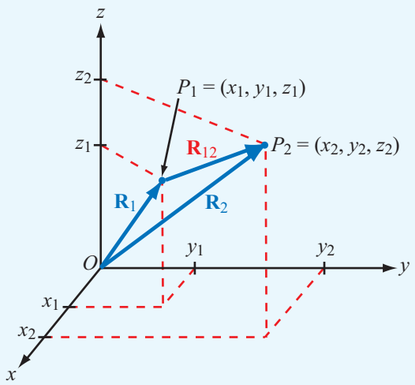


Figure 3-4 Distance vector $\mathbf{R}_{12} = \overrightarrow{P_1P_2} = \mathbf{R}_2 - \mathbf{R}_1$, where $\mathbf{R}_1$ and $\mathbf{R}_2$ are the position vectors of points P_1 and P_2, respectively.

$$\mathbf{R}_2 = \overrightarrow{OP_2} = \hat{\mathbf{x}}x_2 + \hat{\mathbf{y}}y_2 + \hat{\mathbf{z}}z_2, \tag{3.10b}$$

where point O is the origin.

The ***distance vector*** from P_1 to P_2 is defined as

$$\mathbf{R}_{12} = \overrightarrow{P_1P_2} = \mathbf{R}_2 - \mathbf{R}_1$$
$$= \hat{\mathbf{x}}(x_2 - x_1) + \hat{\mathbf{y}}(y_2 - y_1) + \hat{\mathbf{z}}(z_2 - z_1), \tag{3.11}$$

and the distance d between P_1 and P_2 equals the magnitude of $\mathbf{R}_{12}$:

$$d = |\mathbf{R}_{12}| = [(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]^{1/2}. \tag{3.12}$$

Note that the first and second subscripts of $\mathbf{R}_{12}$ denote the locations of its tail and head, respectively (**Fig. 3-4**).

3-1.4 Vector Multiplication

There exist three types of products in vector calculus: the simple product, the scalar (or dot) product, and the vector (or cross) product.

Simple Product

The multiplication of a vector by a scalar is called a ***simple product***. The product of the vector $\mathbf{A} = \hat{\mathbf{a}}A$ by a scalar k results in a vector $\mathbf{B}$ with magnitude $B = kA$ and direction the same as $\mathbf{A}$. That is, $\hat{\mathbf{b}} = \hat{\mathbf{a}}$. In Cartesian coordinates,

$$\mathbf{B} = k\mathbf{A} = \hat{\mathbf{a}}kA = \hat{\mathbf{x}}(kA_x) + \hat{\mathbf{y}}(kA_y) + \hat{\mathbf{z}}(kA_z)$$
$$= \hat{\mathbf{x}}B_x + \hat{\mathbf{y}}B_y + \hat{\mathbf{z}}B_z. \tag{3.13}$$

Scalar or Dot Product

The ***scalar*** (or ***dot***) ***product*** of two co-anchored vectors $\mathbf{A}$ and $\mathbf{B}$, denoted $\mathbf{A}\cdot\mathbf{B}$ and pronounced "A dot B," is defined geometrically as the product of the magnitude of $\mathbf{A}$ and the scalar component of $\mathbf{B}$ along $\mathbf{A}$, or vice versa. Thus,

$$\mathbf{A}\cdot\mathbf{B} = AB\cos\theta_{AB}, \tag{3.14}$$

where θ_{AB} is the angle between $\mathbf{A}$ and $\mathbf{B}$ (**Fig. 3-5**) measured *from* the tail of $\mathbf{A}$ *to* the tail of $\mathbf{B}$. Angle θ_{AB} is assumed to be in the range $0 \le \theta_{AB} \le 180°$. The scalar product of $\mathbf{A}$ and $\mathbf{B}$ yields a scalar whose magnitude is less than or equal to the products of their magnitudes (equality holds when $\theta_{AB} = 0$) and whose sign is positive if $0 < \theta_{AB} < 90°$ and negative if $90° < \theta_{AB} < 180°$. When $\theta_{AB} = 90°$, $\mathbf{A}$ and $\mathbf{B}$ are orthogonal, and their dot product is zero. The quantity $A\cos\theta_{AB}$ is the scalar component of $\mathbf{A}$ along $\mathbf{B}$. Similarly, $B\cos\theta_{BA}$ is the scalar component of $\mathbf{B}$ along $\mathbf{A}$.

The dot product obeys both the commutative and distributive properties of multiplication:

$$\mathbf{A}\cdot\mathbf{B} = \mathbf{B}\cdot\mathbf{A}, \tag{3.15a}$$

(commutative property)

$$\mathbf{A}\cdot(\mathbf{B}+\mathbf{C}) = \mathbf{A}\cdot\mathbf{B} + \mathbf{A}\cdot\mathbf{C}. \tag{3.15b}$$

(distributive property)

The commutative property follows from Eq. (3.14) and the fact that $\theta_{AB} = \theta_{BA}$. The distributive property expresses the fact that the scalar component of the sum of two vectors along a third one equals the sum of their respective scalar components.

The dot product of a vector with itself gives

$$\mathbf{A}\cdot\mathbf{A} = |\mathbf{A}|^2 = A^2, \tag{3.16}$$

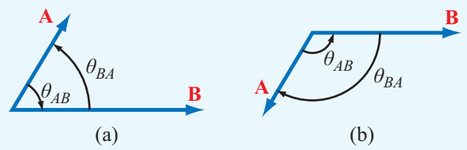


Figure 3-5 The angle θ_{AB} is the angle between $\mathbf{A}$ and $\mathbf{B}$, measured from $\mathbf{A}$ to $\mathbf{B}$ between vector tails. The dot product is positive if $0 \le \theta_{AB} < 90°$, as in (a), and it is negative if $90° < \theta_{AB} \le 180°$, as in (b).

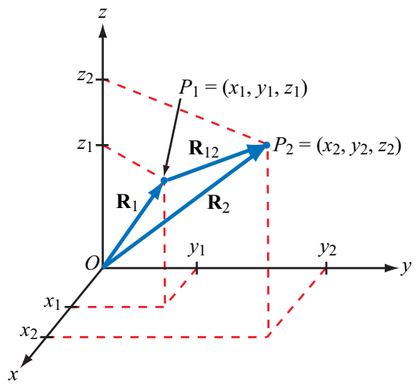


图3-4 距离向量$\mathbf{R}_{12} = \overrightarrow{P_1P_2} = \mathbf{R}_2 - \mathbf{R}_1$，其中$\mathbf{R}_1$和$\mathbf{R}_2$分别是点$P_1$和$P_2$的位置向量。

$$\mathbf{R}_2 = \overrightarrow{OP_2} = \hat{\mathbf{x}}x_2 + \hat{\mathbf{y}}y_2 + \hat{\mathbf{z}}z_2, \tag{3.10b}$$

其中 O 点是原点。

从 P_1 到 P_2 的距离向量定义为

$$\mathbf{R}_{12} = \overrightarrow{P_1P_2} = \mathbf{R}_2 - \mathbf{R}_1$$
$$= \hat{\mathbf{x}}(x_2 - x_1) + \hat{\mathbf{y}}(y_2 - y_1) + \hat{\mathbf{z}}(z_2 - z_1), \tag{3.11}$$

P_1 和 P_2 之间的距离 d 等于 R_{12} 的大小：

$$d = |\mathbf{R}_{12}| = [(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]^{1/2}. \tag{3.12}$$

请注意，R_{12} 的第一个和第二个下标分别表示其尾部和头部的位置（图 3-4）。

3-1.4 向量乘法

向量微积分中存在三种类型的乘积：简单乘积、标量（或点）乘积和向量（或叉）乘积。

标量与矢量相乘

矢量与实标量 k 相乘得到 B=kA，其模为 |B|=|k||A|。当 A≠0 时，k>0则同向，k<0则反向；k=0时为零矢量，无确定方向。原书关于模和方向的表述仅适用于 k>0。直角坐标下：

$$\mathbf{B} = k\mathbf{A} = \hat{\mathbf{a}}kA = \hat{\mathbf{x}}(kA_x) + \hat{\mathbf{y}}(kA_y) + \hat{\mathbf{z}}(kA_z)$$
$$= \hat{\mathbf{x}}B_x + \hat{\mathbf{y}}B_y + \hat{\mathbf{z}}B_z. \tag{3.13}$$

标量或点积

两个共同锚定向量 A 和 B 的标量（或点）积，表示为 A · B 并发音为 "A 点 B"，在几何上定义为 A 的大小与 B 沿 A 的标量分量的乘积，反之亦然。因此，

$$\mathbf{A}\cdot\mathbf{B} = AB\cos\theta_{AB}, \tag{3.14}$$

对非零矢量 A、B，夹角 θ_{AB} 是将两矢量平移到共同起点后，两方向间的角，取值为0°至180°。点积是标量，且 |A·B|≤|A||B|；同向或反向平行时绝对值取等号。

当$0° \le \theta_{AB} < 90°$时点积为正；$90° < \theta_{AB} \le 180°$时为负；$\theta_{AB} = 90°$时两矢量正交，点积为零。A $\cos\theta_{AB}$ 是 A 沿 B 方向的标量分量，B $\cos\theta_{BA}$ 是 B 沿 A 方向的标量分量。

点积遵循乘法的交换律和分配律：

$$\mathbf{A}\cdot\mathbf{B} = \mathbf{B}\cdot\mathbf{A}, \tag{3.15a}$$

(交换律)

$$\mathbf{A}\cdot(\mathbf{B}+\mathbf{C}) = \mathbf{A}\cdot\mathbf{B} + \mathbf{A}\cdot\mathbf{C}. \tag{3.15b}$$

(分配财产)

交换律由等式得出。(3.14) 以及 $\theta_{AB} = \theta_{BA}$ 的事实。分配性质表达了这样一个事实：两个向量之和沿第三个向量的标量分量等于它们各自的标量分量之和。

向量与其自身的点积给出

$$\mathbf{A}\cdot\mathbf{A} = |\mathbf{A}|^2 = A^2, \tag{3.16}$$

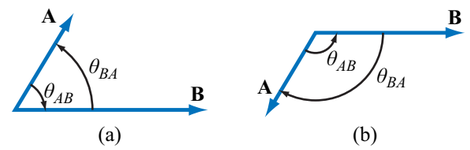


图3-5 矢量 A、B 的夹角。两矢量取共同起点后测量该角；(a) 夹角小于90°，点积为正；(b) 夹角大于90°，点积为负。

which implies that

$$A = |\mathbf{A}| = \sqrt[+]{\mathbf{A}\cdot\mathbf{A}}. \tag{3.17}$$

Also, θ_{AB} can be determined from

$$\theta_{AB} = \cos^{-1}\left[\frac{\mathbf{A}\cdot\mathbf{B}}{\sqrt[+]{\mathbf{A}\cdot\mathbf{A}}\ \sqrt[+]{\mathbf{B}\cdot\mathbf{B}}}\right]. \tag{3.18}$$

Since the base vectors $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$ are each orthogonal to the other two, it follows that

$$\hat{\mathbf{x}}\cdot\hat{\mathbf{x}} = \hat{\mathbf{y}}\cdot\hat{\mathbf{y}} = \hat{\mathbf{z}}\cdot\hat{\mathbf{z}} = 1, \tag{3.19a}$$

$$\hat{\mathbf{x}}\cdot\hat{\mathbf{y}} = \hat{\mathbf{y}}\cdot\hat{\mathbf{z}} = \hat{\mathbf{z}}\cdot\hat{\mathbf{x}} = 0. \tag{3.19b}$$

If $\mathbf{A} = (A_x, A_y, A_z)$ and $\mathbf{B} = (B_x, B_y, B_z)$, then

$$\mathbf{A}\cdot\mathbf{B} = (\hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z)\cdot(\hat{\mathbf{x}}B_x + \hat{\mathbf{y}}B_y + \hat{\mathbf{z}}B_z). \tag{3.20}$$

Use of Eqs. (3.19a) and (3.19b) in Eq. (3.20) leads to

$$\mathbf{A}\cdot\mathbf{B} = A_xB_x + A_yB_y + A_zB_z. \tag{3.21}$$

Vector or Cross Product

The ***vector*** (or ***cross***) ***product*** of two vectors **A** and **B**, denoted $\mathbf{A}\times\mathbf{B}$ and pronounced "*A* cross *B*," yields a vector defined as

$$\mathbf{A}\times\mathbf{B} = \hat{\mathbf{n}}\,AB\sin\theta_{AB}, \tag{3.22}$$

where $\hat{\mathbf{n}}$ is a *unit vector normal to the plane containing* **A** *and* **B** (**Fig. 3-6(a)**). The magnitude of the cross product, $AB|\sin\theta_{AB}|$, equals the area of the parallelogram defined by the two vectors. The direction of $\hat{\mathbf{n}}$ is governed by the ***right-hand rule*** (**Fig. 3-6(b)**): $\hat{\mathbf{n}}$ points in the direction of the right thumb when the fingers rotate from **A** to **B** through the angle θ_{AB}. Note that, since $\hat{\mathbf{n}}$ is perpendicular to the plane containing **A** and **B**, $\mathbf{A}\times\mathbf{B}$ is perpendicular to both vectors **A** and **B**.

The cross product is anticommutative and distributive:

$$\mathbf{A}\times\mathbf{B} = -\mathbf{B}\times\mathbf{A} \quad \textbf{(anticommutative)}. \tag{3.23a}$$

The anticommutative property follows from the application of the right-hand rule to determine $\hat{\mathbf{n}}$. The distributive property follows from the fact that the area of the parallelogram formed by **A** and $(\mathbf{B}+\mathbf{C})$ equals the sum of those formed by (**A** and **B**) and (**A** and **C**):

$$\mathbf{A}\times(\mathbf{B}+\mathbf{C}) = \mathbf{A}\times\mathbf{B} + \mathbf{A}\times\mathbf{C}, \quad \textbf{(distributive)} \tag{3.23b}$$

The cross product of a vector with itself vanishes. That is,

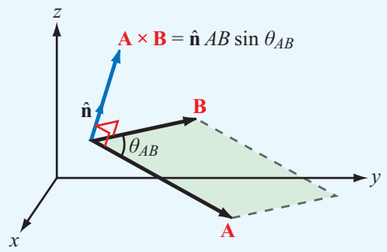
$$\mathbf{A}\times\mathbf{A} = 0. \tag{3.24}$$

From the definition of the cross product given by Eq. (3.22), it is easy to verify that the base vectors $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$ of the Cartesian coordinate system obey the right-hand cyclic relations:

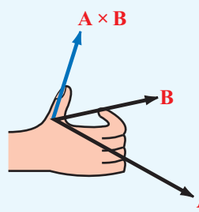
$$\hat{\mathbf{x}}\times\hat{\mathbf{y}} = \hat{\mathbf{z}}, \qquad \hat{\mathbf{y}}\times\hat{\mathbf{z}} = \hat{\mathbf{x}}, \qquad \hat{\mathbf{z}}\times\hat{\mathbf{x}} = \hat{\mathbf{y}}. \tag{3.25}$$

Note the cyclic order (*xyzxyz*...). Also,

$$\hat{\mathbf{x}}\times\hat{\mathbf{x}} = \hat{\mathbf{y}}\times\hat{\mathbf{y}} = \hat{\mathbf{z}}\times\hat{\mathbf{z}} = 0. \tag{3.26}$$



(a) Cross product



(b) Right-hand rule

Figure 3-6 Cross product $\mathbf{A}\times\mathbf{B}$ points in the direction $\hat{\mathbf{n}}$, which is perpendicular to the plane containing **A** and **B** and defined by the right-hand rule.

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这意味着

$$A = |\mathbf{A}| = \sqrt[+]{\mathbf{A}\cdot\mathbf{A}}. \tag{3.17}$$

此外，θ_{AB} 可以由下式确定：

$$\theta_{AB} = \cos^{-1}\left[\frac{\mathbf{A}\cdot\mathbf{B}}{\sqrt[+]{\mathbf{A}\cdot\mathbf{A}}\ \sqrt[+]{\mathbf{B}\cdot\mathbf{B}}}\right]. \tag{3.18}$$

由于基向量 x̂、ŷ 和 ẑ 均与其他两个正交，因此可以得出：

$$\hat{\mathbf{x}}\cdot\hat{\mathbf{x}} = \hat{\mathbf{y}}\cdot\hat{\mathbf{y}} = \hat{\mathbf{z}}\cdot\hat{\mathbf{z}} = 1, \tag{3.19a}$$

$$\hat{\mathbf{x}}\cdot\hat{\mathbf{y}} = \hat{\mathbf{y}}\cdot\hat{\mathbf{z}} = \hat{\mathbf{z}}\cdot\hat{\mathbf{x}} = 0. \tag{3.19b}$$

如果 A = (A_x、A_y、A_z) 和 B = (B_x、B_y、B_z)，则

$$\mathbf{A}\cdot\mathbf{B} = (\hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z)\cdot(\hat{\mathbf{x}}B_x + \hat{\mathbf{y}}B_y + \hat{\mathbf{z}}B_z). \tag{3.20}$$

使用等式。式（3.19a）和（3.19b）(3.20) 导致

$$\mathbf{A}\cdot\mathbf{B} = A_xB_x + A_yB_y + A_zB_z. \tag{3.21}$$

向量或叉积

两个向量 A 和 B 的向量（或叉）乘积，表示为 A× B，发音为“A cross B”，产生一个向量，定义为

$$\mathbf{A}\times\mathbf{B} = \hat{\mathbf{n}}\,AB\sin\theta_{AB}, \tag{3.22}$$

其中n̂是垂直于包含A和B的平面的单位向量（图3-6（a））。叉积 AB| sin θ_{AB}| 的大小等于由两个向量定义的平行四边形的面积。 n̂ 的方向遵循右手定则（图 3-6（b））：当手指从 A 旋转到 B 角度 θ_{AB} 时，n̂ 指向右手拇指的方向。请注意，由于 n̂ 垂直于包含 A 和 B 的平面，因此 A× B 垂直于向量 A 和 B。

叉积是反交换的和分配的：

$$\mathbf{A}\times\mathbf{B} = -\mathbf{B}\times\mathbf{A} \quad \textbf{(anticommutative)}. \tag{3.23a}$$

反交换性质是根据　右手定则的应用来确定的。分配律是根据平行四边形的面积形成的事实得出的

$$\mathbf{A}\times(\mathbf{B}+\mathbf{C}) = \mathbf{A}\times\mathbf{B} + \mathbf{A}\times\mathbf{C}, \quad \textbf{(distributive)} \tag{3.23b}$$

向量与其自身的叉积消失。那是，

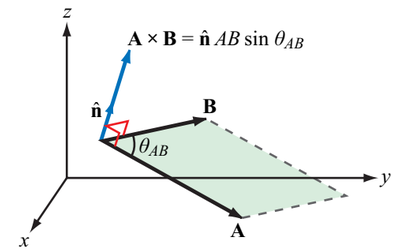
$$\mathbf{A}\times\mathbf{A} = 0. \tag{3.24}$$

根据方程给出的叉积的定义：根据(3.22)式，容易验证笛卡尔坐标系的基向量x̂、ŷ和ẑ服从右手循环关系：

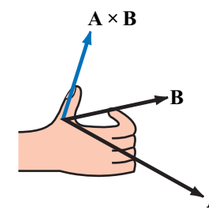
$$\hat{\mathbf{x}}\times\hat{\mathbf{y}} = \hat{\mathbf{z}}, \qquad \hat{\mathbf{y}}\times\hat{\mathbf{z}} = \hat{\mathbf{x}}, \qquad \hat{\mathbf{z}}\times\hat{\mathbf{x}} = \hat{\mathbf{y}}. \tag{3.25}$$

注意循环顺序 (xyzxyz ...)。还，

$$\hat{\mathbf{x}}\times\hat{\mathbf{x}} = \hat{\mathbf{y}}\times\hat{\mathbf{y}} = \hat{\mathbf{z}}\times\hat{\mathbf{z}} = 0. \tag{3.26}$$



(a) Cross product



(b) Right-hand rule

图 3-6 叉积 A × B 指向 n̂ 方向，该方向垂直于包含 A 和 B 的平面，并由右手定则定义。

If $\mathbf{A} = (A_x, A_y, A_z)$ and $\mathbf{B} = (B_x, B_y, B_z)$, then use of Eqs. (3.25) and (3.26) leads to

$$\begin{aligned}\mathbf{A}\times\mathbf{B} &= (\hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z)\times(\hat{\mathbf{x}}B_x + \hat{\mathbf{y}}B_y + \hat{\mathbf{z}}B_z)\\ &= \hat{\mathbf{x}}(A_yB_z - A_zB_y) + \hat{\mathbf{y}}(A_zB_x - A_xB_z)\\ &\quad + \hat{\mathbf{z}}(A_xB_y - A_yB_x). \qquad (3.27)\end{aligned}$$

The cyclical form of the result given by Eq. (3.27) allows us to express the cross product in the form of a determinant:

$$\mathbf{A}\times\mathbf{B} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}. \qquad (3.28)$$

Example 3-1: Vectors and Angles

In Cartesian coordinates, vector **A** points from the origin to point $P_1 = (2,3,3)$, and vector **B** is directed from P_1 to point $P_2 = (1,-2,2)$. Find:

(a) vector **A**, its magnitude A, and unit vector $\hat{\mathbf{a}}$,
(b) the angle between **A** and the y axis,
(c) vector **B**,
(d) the angle θ_{AB} between **A** and **B**, and
(e) perpendicular distance from the origin to vector **B**.

Solution: **(a)** Vector **A** is given by the position vector of $P_1 = (2,3,3)$ (Fig. 3-7). Thus,

$$\mathbf{A} = \hat{\mathbf{x}}2 + \hat{\mathbf{y}}3 + \hat{\mathbf{z}}3,$$

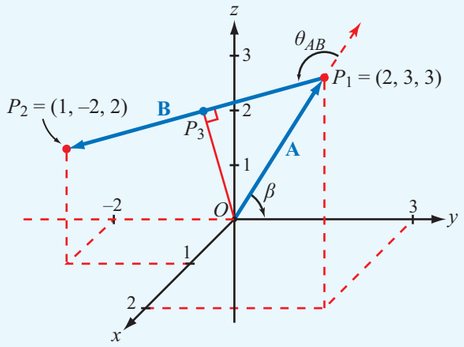


Figure 3-7 Geometry of Example 3-1.

$$A = |\mathbf{A}| = \sqrt{2^2 + 3^2 + 3^2} = \sqrt{22},$$

$$\hat{\mathbf{a}} = \frac{\mathbf{A}}{A} = (\hat{\mathbf{x}}2 + \hat{\mathbf{y}}3 + \hat{\mathbf{z}}3)/\sqrt{22}.$$

(b) The angle β between **A** and the y axis is obtained from

$$\mathbf{A}\cdot\hat{\mathbf{y}} = |\mathbf{A}||\hat{\mathbf{y}}|\cos\beta = A\cos\beta,$$

or

$$\beta = \cos^{-1}\left(\frac{\mathbf{A}\cdot\hat{\mathbf{y}}}{A}\right) = \cos^{-1}\left(\frac{3}{\sqrt{22}}\right) = 50.2°.$$

(c)

$$\mathbf{B} = \hat{\mathbf{x}}(1-2) + \hat{\mathbf{y}}(-2-3) + \hat{\mathbf{z}}(2-3) = -\hat{\mathbf{x}} - \hat{\mathbf{y}}5 - \hat{\mathbf{z}}.$$

(d)

$$\theta_{AB} = \cos^{-1}\left[\frac{\mathbf{A}\cdot\mathbf{B}}{|\mathbf{A}||\mathbf{B}|}\right] = \cos^{-1}\left[\frac{(-2-15-3)}{\sqrt{22}\,\sqrt{27}}\right] = 145.1°.$$

(e) The perpendicular distance between the origin and vector **B** is the distance $|\overrightarrow{OP_3}|$ shown in Fig. 3-7. From right triangle OP_1P_3,

$$\begin{aligned}|\overrightarrow{OP_3}| &= |\mathbf{A}|\sin(180° - \theta_{AB})\\ &= \sqrt{22}\sin(180° - 145.1°) = 2.68.\end{aligned}$$

Example 3-2: Cross Product

Given vectors $\mathbf{A} = \hat{\mathbf{x}}2 - \hat{\mathbf{y}} + \hat{\mathbf{z}}3$ and $\mathbf{B} = \hat{\mathbf{y}}2 - \hat{\mathbf{z}}3$, compute (a) $\mathbf{A}\times\mathbf{B}$, (b) $\hat{\mathbf{y}}\times\mathbf{B}$, and (c) $(\hat{\mathbf{y}}\times\mathbf{B})\cdot\mathbf{A}$.

Solution: **(a)** Application of Eq. (3.28) gives

$$\begin{aligned}\mathbf{A}\times\mathbf{B} &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ 2 & -1 & 3 \\ 0 & 2 & -3 \end{vmatrix}\\ &= \hat{\mathbf{x}}((-1)\times(-3) - 3\times 2) - \hat{\mathbf{y}}(2\times(-3) - 3\times 0)\\ &\quad + \hat{\mathbf{z}}(2\times 2 - (-1\times 0))\\ &= -\hat{\mathbf{x}}3 + \hat{\mathbf{y}}6 + \hat{\mathbf{z}}4.\end{aligned}$$

(b) $\hat{\mathbf{y}}\times\mathbf{B} = \hat{\mathbf{y}}\times(\hat{\mathbf{y}}2 - \hat{\mathbf{z}}3) = -\hat{\mathbf{x}}3.$

(c) $(\hat{\mathbf{y}}\times\mathbf{B})\cdot\mathbf{A} = -\hat{\mathbf{x}}3\cdot(\hat{\mathbf{x}}2 - \hat{\mathbf{y}} + \hat{\mathbf{z}}3) = -6.$

Exercise 3-1: Find the distance vector between $P_1 = (1,2,3)$ and $P_2 = (-1,-2,3)$ in Cartesian coordinates.

Answer: $\overrightarrow{P_1P_2} = -\hat{\mathbf{x}}2 - \hat{\mathbf{y}}4$. (See EM.)

如果 A = (A$_x$、A$_y$、A$_z$) 和 B = (B$_x$、B$_y$、B$_z$)，则使用等式： (3.25) 和 (3.26) 导致

$$\begin{aligned}\mathbf{A}\times\mathbf{B} &= (\hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z)\times(\hat{\mathbf{x}}B_x + \hat{\mathbf{y}}B_y + \hat{\mathbf{z}}B_z)\\ &= \hat{\mathbf{x}}(A_yB_z - A_zB_y) + \hat{\mathbf{y}}(A_zB_x - A_xB_z)\\ &\quad + \hat{\mathbf{z}}(A_xB_y - A_yB_x). \qquad (3.27)\end{aligned}$$

方程给出的结果的循环形式。 (3.27) 允许我们以行列式的形式表达叉积：

$$\mathbf{A}\times\mathbf{B} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}. \qquad (3.28)$$

例 3-1：向量和角度

在笛卡尔坐标系中，矢量A从原点指向点P$_1$ = (2,3,3)，矢量B从P$_1$指向点P$_2$ = (1,−2,2)。寻找：

(a) 向量 A、其大小 A 和单位向量 â，(b) A 与 y 轴之间的角度，(c) 向量 B，(d) A 与 B 之间的角度 θ_{AB}，以及 (e) 从原点到向量 B 的垂直距离。

解： (a)向量A由P$_1$ = (2,3,3) (的位置向量给出。 3-7)。因此，

$$\mathbf{A} = \hat{\mathbf{x}}2 + \hat{\mathbf{y}}3 + \hat{\mathbf{z}}3,$$

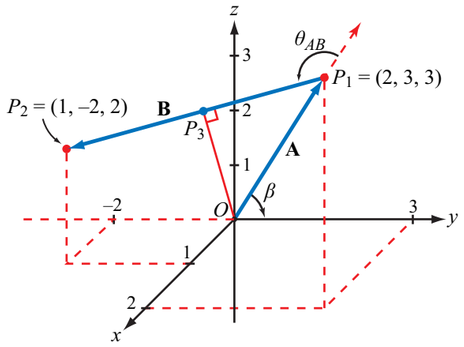


图 3-7 示例 3-1 的几何形状。

$$A = |\mathbf{A}| = \sqrt{2^2 + 3^2 + 3^2} = \sqrt{22},$$

$$\hat{\mathbf{a}} = \frac{\mathbf{A}}{A} = (\hat{\mathbf{x}}2 + \hat{\mathbf{y}}3 + \hat{\mathbf{z}}3)/\sqrt{22}.$$

(b) A 与 y 轴之间的角度 β 由下式获得

$$\mathbf{A}\cdot\hat{\mathbf{y}} = |\mathbf{A}||\hat{\mathbf{y}}|\cos\beta = A\cos\beta,$$

或者

$$\beta = \cos^{-1}\left(\frac{\mathbf{A}\cdot\hat{\mathbf{y}}}{A}\right) = \cos^{-1}\left(\frac{3}{\sqrt{22}}\right) = 50.2°.$$

(c)

$$\mathbf{B} = \hat{\mathbf{x}}(1-2) + \hat{\mathbf{y}}(-2-3) + \hat{\mathbf{z}}(2-3) = -\hat{\mathbf{x}} - \hat{\mathbf{y}}5 - \hat{\mathbf{z}}.$$

(d)

$$\theta_{AB} = \cos^{-1}\left[\frac{\mathbf{A}\cdot\mathbf{B}}{|\mathbf{A}||\mathbf{B}|}\right] = \cos^{-1}\left[\frac{(-2-15-3)}{\sqrt{22}\,\sqrt{27}}\right] = 145.1°.$$

(e) 原点与矢量 B 之间的垂直距离为图 3-7 所示的距离 $|\overrightarrow{OP_3}|$。由直角三角形 OP$_1$P$_3$，

$$\begin{aligned}|\overrightarrow{OP_3}| &= |\mathbf{A}|\sin(180° - \theta_{AB})\\ &= \sqrt{22}\sin(180° - 145.1°) = 2.68.\end{aligned}$$

例 3-2：叉积

给定向量 A = x̂2 − ŷ + ẑ3 和 B = ŷ2 − ẑ3，计算 (a) A× B、(b) ŷ × B 和 (c) (ŷ× B) · A。

解决方案： （a）应用方程。 (3.28) 给出

$$\begin{aligned}\mathbf{A}\times\mathbf{B} &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ 2 & -1 & 3 \\ 0 & 2 & -3 \end{vmatrix}\\ &= \hat{\mathbf{x}}((-1)\times(-3) - 3\times 2) - \hat{\mathbf{y}}(2\times(-3) - 3\times 0)\\ &\quad + \hat{\mathbf{z}}(2\times 2 - (-1\times 0))\\ &= -\hat{\mathbf{x}}3 + \hat{\mathbf{y}}6 + \hat{\mathbf{z}}4.\end{aligned}$$

(b) ŷ× B = ŷ× (ŷ2 − ẑ3) = −x̂3。

(c) (ŷ × B) · A = −x̂3 ·(x̂2 − ŷ + ẑ3) = −6。

练习 3-1：求笛卡尔坐标中 P$_1$ = (1, 2, 3) 和 P$_2$ = (−1 , −2, 3) 之间的距离向量。

Answer: $\overrightarrow{P_1P_2} = -\hat{\mathbf{x}}2 - \hat{\mathbf{y}}4$. (See EM.)